Physical Models of Biological Systems
Quick Reference

Course 77972

2026–2027

1 Model formulation and validation

1.1 Modelling cycle

1. Formulate: specify question, variables, processes, assumptions, and observations.
2. Solve: derive consequences analytically where possible; use computation for remaining behaviour.
3. Validate: compare with observations, limiting cases, units, and invariants; revise assumptions when needed.

1.2 Inference from dynamics

For a model $\dot{x} = f(x, \theta)$, fitting estimates parameters θ from measured trajectories. Count independent features rather than raw data points; correlated measurements may contain less independent information. Structural identifiability asks whether distinct parameter values can produce exactly the same ideal observable data.

2 Population turnover

2.1 Input, output, and steady state

$$\frac{dW}{dt} = Q_{in} - Q_{out}, \qquad \text{steady state: } Q_{in} = Q_{out}.$$

A steady amount does not determine the magnitude of the balanced fluxes. For constant input and first-order removal,

$$\frac{dW}{dt} = Q_{in} - \kappa W, \qquad W^* = \frac{Q_{in}}{\kappa}, \qquad W(t) - W^* = [W(0) - W^*]e^{-\kappa t}.$$

2.2 Two-stage viral clearance after blocking new infection

With constant infected-cell clearance k_I, virion clearance k_V, production per infected cell γ, and $\beta = \gamma N_{I0}$:

$$\dot{N}_I = -k_I N_I, \qquad N_I(t) = N_{I0}e^{-k_I t}, \qquad \dot{N}_V = \beta e^{-k_I t} - k_V N_V.$$

General solution, for $k_V \neq k_I$:

$$N_V(t) = N_{V0}e^{-k_V t} + \frac{\beta}{k_V - k_I}(e^{-k_I t} - e^{-k_V t}).$$

If pretreatment viral production and clearance balance, $\beta = k_V N_{V0}$:

$$\frac{N_V(t)}{N_{V0}} = \frac{k_V e^{-k_I t} - k_I e^{-k_V t}}{k_V - k_I}, \qquad \dot{N}_V(0) = 0.$$

Equal-rate limit $k_I = k_V = k$:

$$N_V(t) = (N_{V0} + \beta t)e^{-kt}, \qquad \beta = kN_{V0} \Longrightarrow N_V(t) = N_{V0}(1 + kt)e^{-kt}.$$

This limit also has zero initial slope. Under the steady-state constraint, exchanging k_I and k_V leaves the normalized viral-load curve unchanged; the curve alone therefore cannot assign the two rate constants to their biological processes.

2.3 First-order decay scales

For $N(t) = N_0e^{-kt}$, the half-life is $t_{1/2} = \ln 2/k$. A factor-F decrease over interval Δt corresponds to average apparent rate $r = \ln F/\Delta t$ if summarized by a single exponential; for a multiexponential trajectory this is not necessarily an individual process rate.